SILAS M. FALDE

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SUMMARY

Highly motivated and detail-oriented Data Science junior and researcher at the University of Michigan, with a strong foundation in the programming languages C++ and Python. Proficient in data analysis and software development, with a passion for applying technical skills to a wide variety of problems. Currently, seeking opportunities to both apply and enhance my skills and knowledge to contribute to innovative projects and teams.

EDUCATION

University of Michigan School of Engineering — **Ann Arbor, MI**
Senior B.E.S. in Data Science, GPA: 3.09/4.00 — *August 2023 – December 2026*

Relevant Courses: Machine Learning, Artificial Intelligence, Computer Vision, Computational Modelling of Complex Systems, Database Management, Regression Analysis, Probability and Statistics, Bayesian Data Analysis, Data Structures and Algorithms

Western High School — **Parma, MI**
GPA: 3.963 — *August 2019 – May 2023*

- Men's Varsity Soccer – Captain 1 year, Member 2 years
- Select Choir – Member 3 years

RESEARCH EXPERIENCE

University of Michigan Department of Epidemiology — **Ann Arbor, MI**
Research Assistant – Computational Lead — *September 2025 – May 2026*

- Led development of a Python pipeline to classify collective-violence narratives using prompt-based large language model inference and structured response processing.
- Built reusable preprocessing utilities for NVDRS-style text, demographic derivation, path management, and annotation-round data handling.
- Designed human annotation workflows and measured inter-annotator agreement with Cohen's kappa and Krippendorff's alpha to resolve disagreements and assess model performance.

University of Michigan Department of Epidemiology — **Ann Arbor, MI**
Research Assistant — *September 2024 – May 2025*

- Earned 3rd place in the CDC Youth Mental Health: Novel Variables Data Challenge, placing in roughly the top 1% of entries.
- Developed Python pipelines on a high-performance computing cluster to clean, normalize, and engineer features from more than 400,000 suicide-victim narratives.
- Used large language models to extract structured trends from narrative text and evaluated model-assisted outputs with classification metrics and error analysis.
- Documented findings with reproducible analyses, reports, and visualizations for the research team and publication.

WORK EXPERIENCE

Ally Financial, Inc. — **Detroit, MI**
Intern, Artificial Intelligence & Advanced Analytics — *May 2026 – August 2026*

- Prototyped large language model workflows to automate internal tasks and improve efficiency and safety.
- Built Python NLP pipelines to evaluate PII masking and redaction model performance for sensitive company documents.
- Enhanced a data dictionary migration tool by adding batch processing, smarter metadata selection, and a streamlined interface to improve usability and efficiency.

Ally Financial, Inc. — **Detroit, MI**
Intern, Data Management — *May 2025 – August 2025*

- Created data dictionaries and lineage documentation in Python to improve data governance and compliance.
- Added data quality checks to existing data pipelines with Python and SQL to improve data integrity and reliability.
- Developed a Power BI data availability dashboard to monitor refresh rates and availability for critical data assets.

Michigan Soccer Referee Association **Ann Arbor, MI**
Certified Grassroots Soccer Referee *March 2022 – Present*

- Officiating on a referee team to create a safe, fair, and fun environment for players, coaches, and spectators.
- Applying knowledge of the rules of soccer to facilitate the sport.

Center Lake Fitness **Jackson, MI**
Groundskeeper *May 2023 – August 2023*

- Mowed, trimmed, and landscaped the grounds to maintain a clean and safe environment for patrons.
- Conducted daily care for farm animals including feeding, cleaning, monitoring animal health, and maintaining facilities.

PROJECTS

Home Server Infrastructure **Ann Arbor, MI**
Systems Developer *2024 – Present*

- Designed and operate a single-host Ubuntu self-hosting platform using Docker Compose stacks for ingress, media, cloud storage, AI services, monitoring, automation, and game hosting.
- Automated media workflows with Bash and Python for queueing, GPU-accelerated HEVC encoding, staging, metadata-safe publishing, library audits, and failure-aware batch processing.
- Configured Cloudflare Tunnel and Caddy ingress, Tailscale private administration, systemd scheduling, service health checks, and documented upgrade, backup, recovery, and security procedures.

silasfalde.net **Ann Arbor, MI**
Web Developer *2024 – Present*

- Built and maintain a responsive static personal website with modular HTML, CSS, and JavaScript for resume, research, project, education, and photography pages.
- Developed browser-side navigation, responsive layouts, photography galleries, asset loading, and reusable site-content modules without a Node build system.
- Automated resume publishing by parsing LaTeX section files into browser-friendly JSON, and generated photography event manifests from organized image folders.

PyNVDRS **Ann Arbor, MI**
Package Developer *2025 – 2026*

- Refactored reusable utilities from NVDRS-style epidemiology projects into a typed, installable Python package with editable-install support.
- Organized shared modules for paths, narrative text cleaning, demographic preprocessing, annotation workflows, and optional GPT-assisted inference.
- Added import and path tests, package metadata, optional dependency groups, and continuous integration to support migration from research scripts.

FPS PvP Dynamics **Ann Arbor, MI**
Project Developer *March 2026 – April 2026*

- Built a configurable **agent-based simulation** in Python to model player movement, combat, objectives, respawning, and stochastic per-tick match dynamics.
- Developed trace capture, metrics export, visualization, and automated tests for debugging and validating simulation behavior.
- Ran parallel parameter sweeps, replication studies, **Latin hypercube sampling**, and **Sobol sensitivity analysis** to quantify the effect of model parameters.

Photo Processing Helper **Ann Arbor, MI**
Project Developer *Jan 2026 - Present*

- Built a modular Python CLI and library for photo framing, collage generation, panorama stitching, and Nikon RAW-to-JPG matching across large image folders.
- Implemented OpenCV, NumPy, Pillow, and rawpy pipelines for resizing, cropping, alignment, blending, and exact-dimension output while preserving high-bit-depth detail.
- Added tests and validation for file discovery, output integrity, stitching behavior, and repeatable batch processing.

Predicting Survival in the ICU **Ann Arbor, MI**
Project Developer *January 2025 – March 2025*

- Developed logistic-regression, convolutional, and recurrent-neural-network models in Scikit-learn and PyTorch to predict ICU mortality from clinical time-series and static features.

- Built reproducible data loaders and preprocessing for missing values, temporal observations, and engineered clinical features, with configuration-driven training and evaluation.
- Evaluated models with AUROC, visualization, and automated tests covering CNN, transformer, and output behavior.

Capuchin Audio Classification **Ann Arbor, MI**
Machine Learning Developer *Course Project*

- Built an end-to-end PyTorch pipeline to detect capuchin monkey calls in forest recordings by loading, resampling, and batching variable-length audio.
- Converted recordings to padded spectrogram tensors and trained a convolutional neural network with GPU support, checkpointing, and TensorBoard loss tracking.
- Ran inference over field recordings and exported per-recording call counts to CSV for downstream analysis.

Vaccine Uptake Prediction **Ann Arbor, MI**
Data Science Developer *Course Project*

- Built a reproducible Python pipeline to predict multilabel vaccine uptake from NHFS survey data, including preprocessing, fold creation, and model evaluation.
- Compared regularized logistic regression, random forests, and Bayesian hierarchical models using cross-validation, ROC AUC, Brier scores, and calibration plots.
- Organized configuration, data utilities, model utilities, and evaluation code into a modular source package with a notebook-driven workflow.

Image Classification and Transfer Learning **Ann Arbor, MI**
Computer Vision Developer *Course Project*

- Built a complete Python/PyTorch image-classification workflow with dataset preparation, CNN and Vision Transformer models, training scripts, and challenge inference.
- Compared custom CNNs, transfer-learning source and target models, and Vision Transformers, with validation AUROC reported around 0.97 for CNN experiments.
- Added automated tests for datasets, image standardization, CNNs, ViTs, and generated outputs to validate the training and evaluation pipeline.

Sequence Learning and Clustering **Ann Arbor, MI**
Machine Learning Developer *Course Project*

- Implemented and evaluated recurrent neural networks and self-attention models for sequence prediction, achieving a reported test AUROC of 0.803.
- Implemented k-means and hierarchical clustering experiments, comparing initialization strategies and objective values across runs.
- Organized reproducible training configurations, checkpointed models, visualizations, and written analyses across a Python source tree.

Probabilistic Models and Autoencoders **Ann Arbor, MI**
Machine Learning Developer *Course Project*

- Implemented Gaussian mixture model training with expectation-maximization and analyzed convergence and mixture assignments in Python.
- Built experiments for Bayesian networks and autoencoders, including denoising and regularized reconstruction models with saved checkpoints.
- Used numerical modeling, visualization, and LaTeX-based technical reporting to compare probabilistic and representation-learning methods.

Model-Based and Model-Free Reinforcement Learning **Ann Arbor, MI**
Reinforcement Learning Developer *Course Project*

- Implemented Active ADP, a model-based reinforcement-learning agent that learned transition and reward models and planned with value iteration.
- Implemented Q-learning with optimistic exploration and compared model-based and model-free learning in deterministic and stochastic Gymnasium FrozenLake environments.
- Built evaluation, learning-curve, video-recording, and reproducibility utilities with controlled random seeds and configurable experiments.

Database Storage and Grace Hash Join **Ann Arbor, MI**
C++ Developer *Course Project*

- Implemented C++ database-storage components for disk pages, records, memory buffers, buckets, and relation management.

- Built the partition phase of a Grace Hash Join using page-level I/O, hash-based bucket placement, buffer flushing, and disk-page tracking.
- Worked across a multi-file C++ codebase with Make-based builds and explicit memory, page-capacity, and record-layout constraints.

Photo Library Compressor **Ann Arbor, MI**
Python Developer *Personal Project*

- Engineered a Python batch pipeline to compress and reorganize large iCloud photo libraries using file size, format, and GPS metadata.
- Preserved EXIF metadata while applying configurable quality steps, routing missing-location images, and tracking processing results in CSV metrics.
- Added timeout handling, staged processing, and failure-tolerant orchestration for long-running HEIC and JPEG media workloads.

Breast Cancer Detection **Ann Arbor, MI**
Machine Learning Developer *Course Project*

- Investigated breast-cancer image classification with Python notebooks covering exploratory data analysis, convolutional neural networks, EfficientNetV2, and Vision Transformer models.
- Built image preprocessing, training, and evaluation experiments with NumPy, pandas, Pillow, Matplotlib, and PyTorch-oriented workflows.
- Compared transfer-learning approaches and documented model behavior through visual analysis and experiment reports.

Image Recognition and Metric Learning **Ann Arbor, MI**
Computer Vision Developer *Course Project*

- Implemented image-classification and metric-learning experiments in Python and PyTorch for a computer-vision recognition assignment.
- Trained separate classification and embedding models, saved learned weights, and analyzed recognition behavior through notebook experiments and reports.
- Applied feature-space learning techniques to compare image similarity and supervised recognition performance.

Computer Vision Panorama Stitching **Ann Arbor, MI**
Computer Vision Developer *Course Project*

- Implemented a Python/OpenCV panorama pipeline for numerical image processing, feature alignment, and composition of multiple photographs.
- Used NumPy representations and geometric transformations to align image content and produce stitched panoramic outputs.
- Documented algorithmic choices and evaluated results on supplied image datasets through a notebook-based workflow.

SKILLS

Programming: Python, C++, Bash, JavaScript, R, SQL

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, spaCy, NLTK

Web: HTML, CSS, JavaScript, responsive design, static-site publishing

Systems: Linux, Docker Compose, systemd, CMake, Make, Git, CI/CD

Software: VS Code, Microsoft Office Suite, Google Workspace, Snowflake, Collibra, Alation, Manta